MiCro: Modeling Cross-Image Semantic Relationship Dependencies for Class-Incremental Semantic Segmentation in Remote Sensing Images

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***Abstract*— Continual learning is an effective way to overcome catastrophic forgetting (CF) in incremental learning for semantic segmentation. The existing continual semantic segmentation (CSS) methods of remote sensing (RS) ignore the semantic relationships among pixels across different images, which will lead to disappointing segmentation results, such as edge pixel misclassification and small object omission. In this article, we propose a framework for modeling cross-image semantic relationship dependencies (MiCro), which aims to learn an interclass separable and intraclass cohesive feature space from the pixel relationships across various images to ensure that learned categories can prevent CF in the incremental process. Specifically, we exploit the relationships among pixels of images in minibatch to construct three losses: 1) cross-image feature relationship distillation (CFRD) loss, which builds a well-structured feature space; 2) cross-image intraclass feature cohesion (CIFC) loss, which is devised to make intraclass features more cohesive; and 3) cross-image class-area weighted cross-entropy (CCWCE) loss, which is mainly employed to inversely weight the proportion of category area in minibatch. The effectiveness of the proposed approach is demonstrated by extensive experiments on three RS semantic segmentation datasets from ISPRS Vaihingen, ISPRS Potsdam, and iSAID. MiCro is superior to the current most advanced methods in most incremental settings, especially improving mIoU by 11.59% on ISPRS Vaihingen, 13.17% on ISPRS Potsdam, and 15.01% on iSAID in the most difficult incremental settings, which promotes the CSS to a state-of-the-art (SOTA) level. The code will be available at https://github.com/RongXueE/MiCro.**

***Index Terms*— Catastrophic forgetting (CF), class-incremental learning (CIL), remote sensing (RS), semantic segmentation.**

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NOMENCLATURE

CF	Catastrophic forgetting.
RS	Remote sensing.
CSS	Continual semantic segmentation.
CISS	Class-incremental semantic segmentation.
CIL	Class-incremental learning.
CFRD	Cross-image feature relationship distillation.
CIFC	Cross-image intraclass feature cohesion.
CCWCE	Cross-image class-area weighted cross-entropy.
SOTA	State-of-the-art.
KD	Knowledge distillation.
OPD	Output probability distillation.
IFD	Intermediate feature distillation.

I. INTRODUCTION

THE RS semantic segmentation has produced excellent results with the quick development of deep learning and is frequently employed in applications such as road extraction [4], urban change detection [5], and land cover [6]. However, practical scenarios often face the requirements to handle stream-formatted data [7] and new classes, such as meteorological measurement data and the newly annotated categories in the dataset. In this case, an ideal model should be able to identify new classes while preserving the discriminability of learned classes, which is known as CIL. The most difficult challenge of CIL is CF [8], [9], which means that, when optimizing the model with new categories, the previously obtained knowledge on former classes is severely forgotten.

The existing mainstream CISS methods [1], [2], [3] are based on KD [10] strategy to deal with CF because KD requires neither additional storage nor has access to previous data to preserve learned knowledge. These methods usually include cross-entropy (CE) loss and OPD loss (or their variants). For example, MiB [2] addresses the background semantic shift issue by incorporating the probability of true background into the CE and OPD loss calculations. Optionally, some methods [1], [11], [12] impose constraints on intermediate features, which can be summarized as IFD loss.

Although the above methods can achieve good results, a major limitation is that their optimization of the objective function is based on each pixel in a single image, without explicitly considering the relationship among pixels on different images. As a result, the context and structured nature

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of the task are severely overlooked with this design, which will lead to unsatisfactory segmentation results, especially on edge pixels at the junction of categories and small objects. We summarize the challenges posed by the ignorance of global semantic relationships among cross-image pixels in CISS of RS into the following three aspects.

A. Challenge 1: Insufficient Capture of Structured Spatial Knowledge

Whether in natural scenes or RS scenes, distilling global pairwise relationships [13], [14] or local pixel affinity [15] often performs better than classical pointwise alignment in capturing structured spatial knowledge. Some works [16], [17] have proved the effectiveness of modeling semantic relationships in RS images. Research [18] discusses that a well-structured feature space can be constructed according to the global pixel dependency, which helps the new model to better imitate the structured semantic relations of the old model. However, the existing CISS methods only consider distilling the knowledge from a single image between the old and new models [as illustrated in the left part of Fig. 1(a)]. They ignore the global semantic relationship among cross-image pixels that are valuable for distillation [as shown in the right part of Fig. 1(a)]. Therefore, it will help to learn a feature space that is more conducive to resisting CF if the global semantic relationship among cross-image pixels is modeled.

B. Challenge 2: Insufficient Cohesion of Intraclass Features

CE loss optimized with softmax mainly has two limitations. One is that it independently punishes pixelwise prediction but ignores the relationship among pixels [19]. The other is that, while softmax is used, the loss merely depends on the relative relationship among logits, which prevents it from directly supervising the learned representation [20]. As a result, the existing CISS methods that apply such design lead to the learned features usually interclass separability but lack of intraclass cohesion, as shown in Fig. 1(b). For dealing with complex data, such as RS, the features learned by the model are expected to be not only interclass separable but more importantly intraclass cohesive. Only by keeping intraclass aggregation, we can have a more robust result for those samples with large intraclass variance. Therefore, we hope to devise a loss function exploiting the cross-image feature pixel relationship to combine CE loss to learn an interclass separable and intraclass cohesive feature space, which can effectively resist CF in the incremental process.

C. Challenge 3: Insufficient Stability of Attention Weights for Each Category

RS images have the characteristics of large differences in object size. As shown in Fig. 1(c), the small objects in yellow color are densely arranged, and each object of this category has fewer pixels than other categories. After multiple downsampling, there are few small object feature pixels left, and the model naturally prefers to optimize the object with a large proportion of pixels. However, the accuracy evaluation includes all categories, which requires the model to perform well in all categories. Moreover, the pixel proportion of a certain class of object in a single image varies greatly, and the pixel proportion of each class of object in a minibatch is relatively closer to the entire dataset, so the stability is stronger. By inversely weighting the proportion of pixels that belong to the same category in the minibatch, more stable attention weights are obtained to better improve small objects and overall performance.

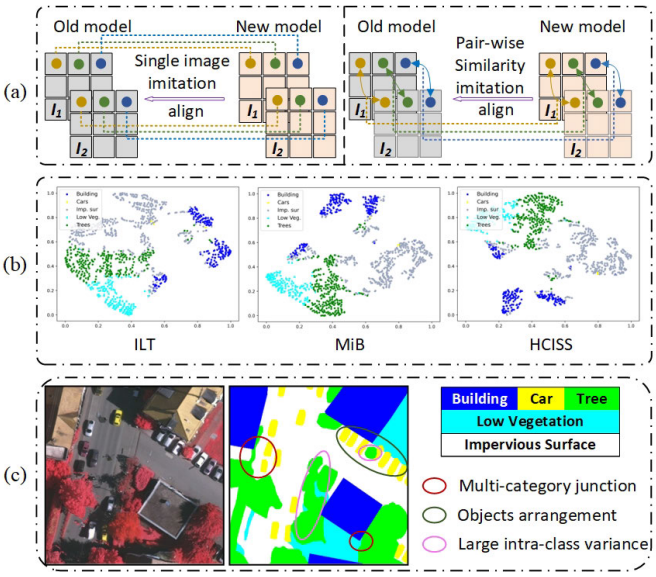


Fig. 1. Three challenges of CISS in RS images. (a) Single image feature imitation and cross-image feature relationship imitation comparison (I_1 and I_2 denote different images, --- represents the imitation operation, and $\leftrightarrow$ indicates pairwise relationship). (b) Features obtained by the existing CISS methods (i.e., ILT [1], MiB [2], and HCISS [3]) trained with CE loss are interclass separable, but intraclass cohesion is insufficient. (c) Complex characteristics of RS images.

In response to the aforementioned three challenges, this article proposes a modeling cross-image semantic relationship dependencies (MiCro) framework. The proposed MiCro includes three losses based on the current training minibatch, which can solve the problem of ignoring the semantic relationship across different image pixels. Specifically, we propose a CFRD loss to model the pairwise similarity between different image embeddings for solving Challenge 1. Furthermore, we present CIFC loss to deal with Challenge 2, which is combined with CE loss (or its variants) to obtain an interclass separable and intraclass cohesive feature space. In addition, to balance the problem of inconsistent attention caused by the difference in the total area of different categories' objects (especially small objects) mentioned in Challenge 3, we propose a CCWCE loss. Fig. 2 illustrates the CSS pipeline between our work and prior research.

To confirm the efficacy and generalization of the proposed strategy, we perform abundant experiments on the ISPRS Vaihingen dataset [21], the ISPRS Potsdam dataset [22], and the iSAID [23] dataset. The experimental results demonstrate that the well-structured, interclass separable, and intraclass cohesive feature space learned with the proposed MiCro framework is conducive to resisting CF. The experiments also approve that our method achieves SOTA compared to recent CSS approaches in most incremental setups. We summarize

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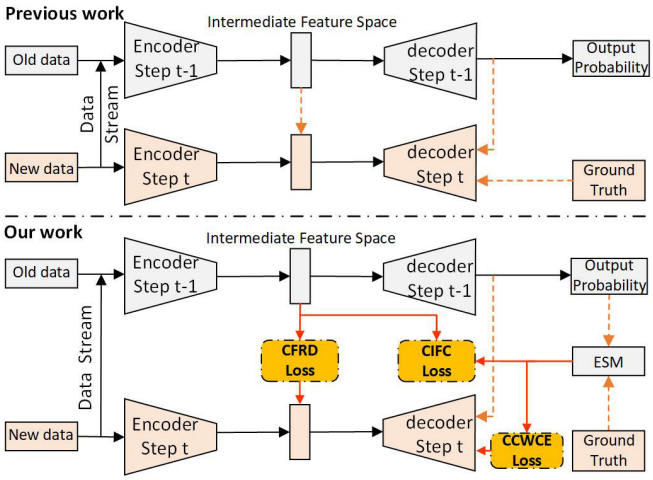


Fig. 2. CSS pipeline between other CSS methods and our work. Orange dotted arrows represent three common losses in CSS tasks. Red solid arrows are associated with the losses of MiCro proposed in this article. Our work aims to construct a well-structured, interclass separable, and intraclass cohesive feature space to prevent CF. The ESM denotes the extended segmentation map acquired by employing pseudolabels. The CFRD, CIFC, and CCWCE indicate CFRD, CIFC, and CCWCE losses, respectively. Best viewed in color.

the main contributions and innovations of this article as follows.

1) This article reveals the importance of modeling cross-image semantic relationships for CISS in RS images. Compared with other CSS approaches that only consider constructing the relationships among pixels of a single image, our method can learn an interclass separable, intraclass cohesive, and well-structured feature space, which is more conducive to preventing CF.
2) We present the CFRD loss to distill the pairwise pixel relationship dependencies across minibatch images from the old model to the new model, which is beneficial to transfer well-structured feature space.
3) The proposed CIFC loss assists CE loss to train the segmentation network and obtains an interclass separable and intraclass cohesive feature space in the incremental process. The learned features are more conducive to resisting CF and also reduce the impact of large intraclass differences in RS images.
4) A CCWCE loss is implemented to guide the network to concentrate more on the learning of categories with fewer pixel proportions (small objects) in the minibatch, thus enhancing the performance of small objects and the overall performance of the model.

II. Related Works

In Section II-A, we provide a brief introduction to semantic segmentation in RS scenes. Following this, we concentrate on the progress of continual learning in both natural and RS scenarios in Section II-B.

A. Semantic Segmentation of Aerial Imagery

The proposal of fully convolutional networks (FCNs) [24] has greatly promoted the development of aerial imagery semantic segmentation, and many impressive works have emerged. These models typically use an encoder–decoder structures and incorporate contextual information about pixel-level features in a variety of ways, such as through pyramids [25], [26], [27], [28], [29] or attention [30], [31], [32], [33], [34], [35], [36], [37] mechanisms. Since the successful application of the attention-based transformer [38] to computer vision, transformer-based approaches have also emerged in the field of RS segmentation. For example, a context transformer is provided by WiCoNet [36] to integrate context information and optionally project it onto local features obtained with a CNN. UNetFormer [39] introduces a transformer-based decoder and builds a UNet-like transformer for the task of real-time urban scene segmentation. RingMo [40] proposes a generative self-supervised foundation model based on the transformer, which achieves a leading level in the downstream tasks of RS semantic segmentation. Although these semantic segmentation methods are effective, they require large datasets to support training, which is frequently expensive to acquire. In addition, while managing tasks requiring sequential learning, they may experience CF.

B. Continual Learning

1) Continual Learning in Natural Scenes: Continual learning aims to equip the model with the capacity to sequentially learn new data without forgetting what it has already learned. Existing methods can be grouped into three categories: data replay, parameter isolation, and regularization methods. The approaches of data replay achieve continual learning from the data perspective by retaining a small number of samples [11], [41], [42], [43] or leveraging the generator to synthesize samples [44], [45], [46] of learned tasks. Parameter isolation, also known as network expansion, implements continual learning at the network structural level. These methods [47], [48], [49], [50], [51], [52], [53] either extend independent network branches for new data or keep a fixed subset of model parameters for all tasks. Regularization techniques can be further separated into prior-focused and data-focused. The former branch [54], [55], [56], [57], [58] focuses on suppressing changes in important parameters related to previously learned task knowledge, while the data-focused methods [1], [2], [12], [59], [60] mainly leverage KD [10] strategy to retain knowledge of learned models. Instead of using the three categories of continual learning methods above separately, there are methods [11], [41], [60], [61], [62] that combine two or three of them.

Though the methods mentioned above can overcome the CF phenomenon in the continual learning process to a certain extent, they all have some defects more or less. The performance of replay-based methods highly depends on the storage capacity and the quantity of the replayed data. Although access to real data can be avoided by employing generative adversarial networks (GANs), continuously training GANs will increase model complexity. The parameter isolation approaches might not be appropriate for a scene with lengthy sequences since their additional parameters will result in expensive computation and significant storage needs as the length of the learning tasks rises. The prior-focused methods

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might be impractical as the amount of parameters grows due to their assumption that each parameter in the network is independent and their requirement to estimate the importance of each parameter.

The KD-based methods are favored because they overcome the above shortcomings and maintain the performance of previous tasks without drastically decreasing. Since learning without forgetting (LWF) [55] initially introduces the KD mechanism into the continual learning concept, many subsequent works [1], [2], [11], [12], [41], [55], [59], [63], [64] have been carried out based on KD, especially in the field of CISS [1], [2], [59], [65], [66]. Incremental learning technique (ILT) [1] is the earliest study that concentrates on the CISS scene without retaining previously seen data, and the authors also apply distillation loss on the intermediate feature. Since ILT presented the CISS task on the standard segmentation datasets, many researchers have focused on this scenario. Modeling the background (MiB) [2] suggests a technique to address the background semantic shift problem existing in the CISS paradigm by incorporating the probability of real background class into the computation of CE and KD loss. PLOP [12] preserves old knowledge through pseudolabel generation and multiscale distillation of spatial features. Yang et al. [66] employ uncertainty-aware contrastive distillation for CISS.

2) Continual Learning in Aerial Imagery: With the vigorous development of incremental learning in natural scenes and its ability to process streaming data in recent years, continual learning has emerged as a blossoming research area in aerial imagery. Compared to the classification task [53], [67], [68], [69] that has attracted extensive focus from researchers, the work on semantic segmentation has received relatively less attention. Tasar et al. [70] first pay attention to CISS by storing samples of learned classes. Rong et al. [3] propose a more challenging CISS setting as they do not access the learned data, while Feng et al. [14] follow with interest in TIL setting without using previous data. This article considers CISS in aerial imagery without accessing the previous data. Different from the above CISS methods that only consider the relationships among pixels in one image, we model semantic dependencies among pixels from minibatch images.

III. Methods

We introduce the proposed MiCro framework in detail in this section. First, we define a symbolic description of CIL in Section III-A. Then, the entire network architecture of the proposed framework is described in Section III-B. After that, we introduce the three proposed components: CFRD loss, CIFC loss, and CCWCE loss in detail in Sections III-C–III-E, respectively. In addition, this article involves many abbreviations. For ease of understanding, we list the main abbreviations and their full names in Nomenclature.

A. Problem Formulation

CIL is an algorithm that divides the entire dataset into different incremental settings, e.g., the ISPRS Vaihingen dataset [21] with five object categories into 4-1 (two steps) and 3-1s (three steps), and then sequentially trains a segmentation model in T steps with less forgetting. The CIL training process consists of several steps or tasks, each of which contains at least one class. Typically, suppose that we have a sequence of supervised learning assignments with T training steps, denoted by $\mathcal{T} = \{1, 2, \ldots, T\}$, and we also have a training dataset denoted D_{tr}, where $D_{\text{tr}} = D_{\text{tr}}^1 \cup D_{\text{tr}}^2 \cup \cdots \cup D_{\text{tr}}^T$ and $D_{\text{tr}}^{t-1} \cap D_{\text{tr}}^t = \emptyset$. Specifically, D_{tr}^t constitutes of a set of pairs (x_i^t, y_i^t), where $x_i^t \in \mathcal{R}^{W\times H\times 3}$ is an image with spatial dimensions of H and W, and the channel dimension is 3. $y_i^t \in \mathcal{R}^{W\times H\times C^t}$ is the corresponding ground-truth (GT) segmentation map, where C^t indicates the current classes' labels. Note that other classes (i.e., previously learned classes $C^{1:t-1}$ or the classes that will be learned $C^{t+1:T}$) are allocated as background class c_{bg}.

Generally, segmentation model S_{Θ^t} comprises a feature extractor $f^t(\cdot)$ for extracting features and a classifier $f_{\text{CLS}}^t(\cdot)$ for obtaining predicted segmentaion maps P^t. The collection of all learnable parameters is denoted by $\Theta^t = \{\theta_l^t\}$, where $l = \{1, 2, \ldots, L\}$ represents the various model layers. $P^t \in \mathcal{R}^{W\times H\times C^{1:t}} = \sigma(O^t)$, where $C^{1:t}$ indicates the categories learned so far, σ means the softmax activation function, and $O^t = (f_{\text{CLS}}(\cdot) \circ f^t(D_{\text{tr}}^t))$ is the output logits map.

B. Network Framework

The proposed MiCro framework is demonstrated in Fig. 3. f_{FCN} denotes the FCN encoder, which maps the input images D_{tr}^t into high-dimensional embeddings $\mathbf{E^t} = f_{\text{FCN}}(D_{\text{tr}}^t) \in \mathcal{R}^{H_f\times W_f\times C_f}$. f_{FCN} can be implemented using any FCN backbone, and ResNet101 [71] is adopted in our experiments. Here, $H_f \times W_f = (H/16) \times (W/16)$ are the feature's height and width, where H and W are the spatial dimensions of input images and $C_f = 2048\text{-}d$ is the channel dimension of embeddings. f_{PROJ} represents the project head, which maps each high-dimensional embedding $\mathbf{e_i^t} \in \mathbf{E^t}$ to a 256-d L_2-normalized feature vector $\mathbf{z_i}$ to calculate $\mathcal{L}_{\text{cifc}}$. f_{PROJ} is attained by two 1×1 convolutional layers with ReLU. Notably, the project head is only used in network training. f_{SEG} indicates segmentation head or decoder, which together with the encoder f_{FCN} forms feature extractor $f^t(\cdot)$. DeepLab-V3 [72] is employed as the segmentation head in this article.

Both the previous model $S_{\Theta^{t-1}}$ and the new model S_{Θ^t} will be provided with training data D_{tr}^t to obtain the features and predicted segmentation maps accordingly. The model $S_{\Theta^{t-1}}$ trained for the last task is frozen, while the current model S_{Θ^t} is learnable. Classically, two loss functions are adopted in CIL to achieve the plasticity-stability balance. The CE loss is employed to gain new categories knowledge, while the OPD loss is applied to distill the output probability distribution from the previous model to the new model for retaining the high performance of the learned classes.

Specifically, the CE loss is formulated as follows:

$$\mathcal{L}_{\text{ce}}\left(x_i^t, y_i^t\right) = -\sum_{c\in C^t} y_i^t[c] \cdot \log\left(P\left(x_i^t; \Theta^t\right)[c]\right) \tag{1}$$

where $P(x_i^t; \Theta^t)[c]$ represents the predicted probability maps and $y_i^t[c]$ indicates the one-hot labels for class c.

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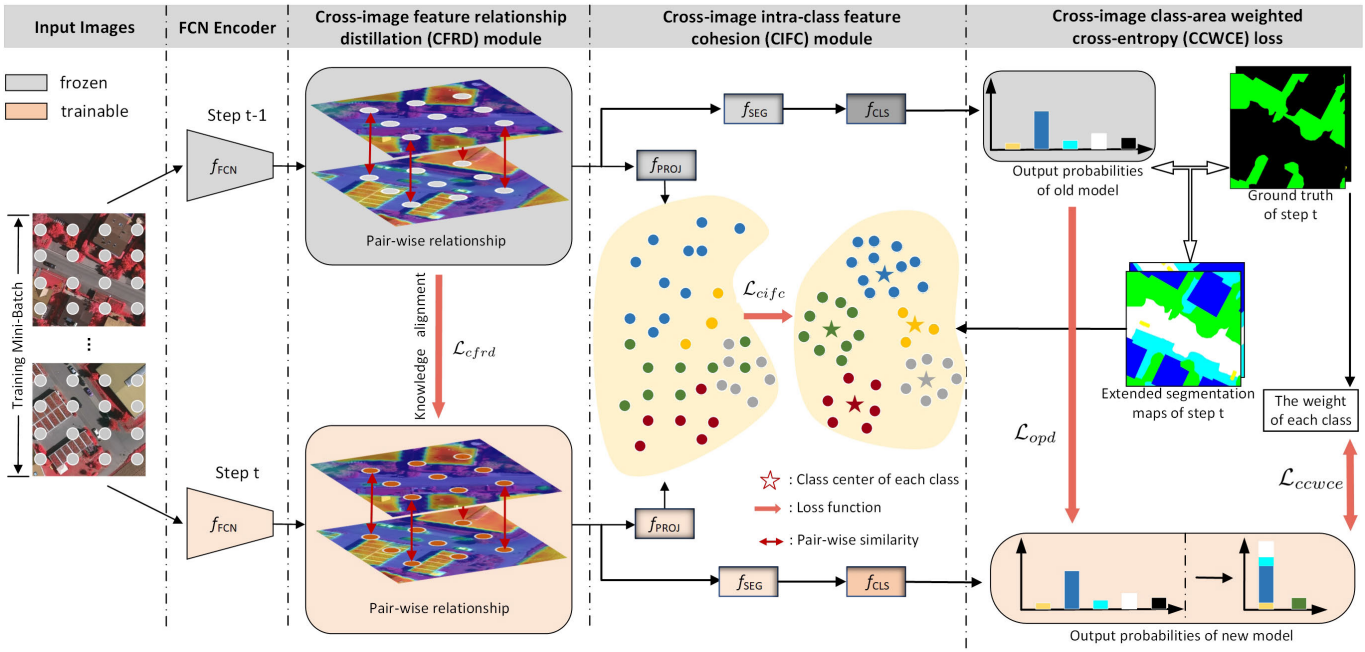


Fig. 3. Overview of the tth incremental step of the proposed MiCro framework. Both the previous model $S_{\Theta^{t-1}}$ and the new model S_{Θ^t} will be provided with training data D_{tr}^t and subjected to joint optimization of the proposed losses to obtain the final predicted segmentation maps. Best viewed in color.

A common option for OPD loss has always been

$$\mathcal{L}_{\text{opd}}\left(x_i^t\right) = -\sum_{c\in\mathcal{Y}^{t-1}} P\left(x_i^t;\Theta^{t-1}\right)[c]\cdot\log\left(P\left(x_i^t;\Theta^t\right)[c]\right) \quad (2)$$

where $\mathcal{Y}^{t-1} = C^{1:t-1}$ denotes old classes, and $P(x_i^t;\Theta^{t-1})[c]$ and $P(x_i^t;\Theta^t)[c]$ are the predicted probability maps of $S_{\Theta^{t-1}}$ and S_{Θ^t}. Minimizing $\mathcal{L}_{\text{opd}}$ is beneficial for the current model to approximate the output distribution of the previous model, thus achieving the purpose of preserving old knowledge.

Alternatively, some methods [1], [3] also perform distillation on intermediate features. Suppose that $e_i^{t-1}, e_i^t \in \mathcal{R}^{H_f\times W_f\times C_f}$ are the intermediate features, respectively, obtained from the ith image by the old and current models. The IFD can be accomplished by the following formula:

$$\mathcal{L}_{ifd} = \frac{1}{H_f\times W_f}\sum_{h_f=1}^{H_f}\sum_{w_f=1}^{W_f}\Omega\mathcal{L}\left(e_i^{t-1}\left(h_f,w_f\right), e_i^t\left(h_f,w_f\right)\right) \quad (3)$$

where Ω is the pixel-level weight mask of the same size as the feature, i.e., binary mask in [73], feature global perception mask in [3], and all 1 mask in [1]. $\mathcal{L}$ can be mean square error (mse) loss or cosine similarity, and (h_f, w_f) represents the corresponding feature pixel.

Although the combination of CE loss, OPD loss, and IFD loss is extensively employed in CISS, they only learn knowledge from the pixels of a single image and neglect the relationships between representations of pixels from different images. Inspired by some recent work on contrastive semantic segmentation [74], [75], we exploit the cross-image embeddings from the current minibatch with N images. Particularly, we propose three cross-image losses, including CFRD loss, CIFC loss, and CCWCE loss, to learn more stable representations, which will enable the model to prevent CF.

The total loss of the proposed framework can be calculated as follows:

$$\mathcal{L}\left(\Theta^t\right) = \frac{1}{|D_{\text{tr}}^t|}\sum_{i=1}^{D_{\text{tr}}^t}\left(\mathcal{L}_{\text{ccwce}}\left(x_i^t, y_i^t\right) + \alpha\mathcal{L}_{\text{opd}}\left(x_i^t\right) + \beta\mathcal{L}_{\text{cfrd}}\left(x_i^t\right) + \gamma\mathcal{L}_{\text{cifc}}\left(x_i^t\right)\right) \quad (4)$$

where α, β, and $\gamma > 0$ are hyperparameters and $|D_{\text{tr}}^t|$ is the number of all images for step t. Note that the CE loss of formula (1) is only used when training task 1.

Algorithm 1 illustrates the CISS pipeline. In the subsections that follow, we will give a thorough introduction to the proposed objective functions.

C. CFRD Loss

The effectiveness and importance of keeping the intermediate features close between the old and new models are demonstrated in some existing CISS methods [1], [3], [12], [14]. However, these methods only consider the pairwise relationships between old and new models in a single image, regardless of modeling semantic relationship dependencies among pixels across different images, as shown in Fig. 4(a). Whether it is a natural scene image or an RS image, there is such a relationship: any pixel pair in two images from the same category (foreground/background) has a greater similarity than pixel pairs from different categories. Therefore, the point-to-point similarity calculation for the features of two images implies the semantic distance relationship between the two feature pixels, which can better capture the structured spatial knowledge. Inspired by some recent contrastive learning approaches [74], [75], we propose the CFRD loss to exploit pixel features beyond a single image by resorting to the current minibatch, as illustrated in Fig. 4(b).

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Algorithm 1 Class-Incremental Learning With MiCro

Input:

$\mathcal{T} = \{1, 2, \cdots, T\}$: sequence segmentation steps;

$D_{tr}^{t} = (x_i^t,\ y_i^t)|_{i=1}^{|D_{tr}^t|}$: training data for step t;

α, β, γ: hyperparameters for the different losses;

Output:

S_{Θ^t}: model optimized by step t;

while t **do**
  Acquire D_{tr}^t for step t;
  if t=1 **then**
    Initialize Θ^1 with the pretrained weights;
    Employ Θ^1 on the training data D_{tr}^1: $P(x_i^1; \Theta^1)$;
    Compute the CE loss: $\mathcal{L}_{\text{ce}}(x_i^1, y_i^1)$ (1);
    Update Θ^1: $\Theta^1 = \text{argmin}_{\{\theta_l^1\}}\mathcal{L}_{\text{ce}}$;
    Save model S_{Θ^t};
  else
    Initialize Θ^t with the old model Θ^{t-1};
    Employ Θ^{t-1} on D_{tr}^t: $P(x_i^t; \Theta^{t-1})$;
    Employ Θ^t on D_{tr}^t: $P(x_i^t; \Theta^t)$;
    Compute the CCWCE loss: $\mathcal{L}_{\text{ccwce}}(x_i^t, y_i^t)$ (16);
    Compute the CFRD loss $\mathcal{L}_{cfrd}(x_i^t, x_j^t)$(6);
    Compute the OPD loss $\mathcal{L}_{\text{opd}}$ (2);
    Compute the CIFC loss $\mathcal{L}_{cifc}$ (11);
    Obtain overall losses: $\mathcal{L}(\Theta^t)$ (4);
    Update learnable parameters: $\Theta^t = \text{argmin}_{\{\theta_l^t\}}\mathcal{L}(\Theta^t)$;
    Save model S_{Θ^t};
  end if
end while
return S_{Θ^t}

Suppose that N is the size of the training minibatch, and the network can obtain N features $\{e_i \in \mathcal{R}^{H_f \times W_f \times C_f}\}|_{i=1}^{N}$ by using the feature extractor $f(\cdot)$. Here, H_f, W_f, and C_f are the height, width, and channel dimensions of e_i, respectively. We reshape the features from $\{e_i \in \mathcal{R}^{H_f \times W_f \times C_f}\}|_{i=1}^{N}$ to $\{e_i \in \mathcal{R}^{A_f \times C_f}\}|_{i=1}^{N}$, where $A_f = H_f \times W_f$. Then, for the ith and the jth images in the minibatch, we can compute the pairwise similarity matrix as follows:

$$S_{ij} = e_i \cdot e_j \in \mathcal{R}^{A_f \times A_f} \tag{5}$$

where $i, j \in \{1, 2, \ldots, N\}$. If the feature pixels of $e_i(h_f, w_f)$ and $e_j(h_f, w_f)$ belong to the same category, then their relationship will be closer, that is, S_{ij} will be larger. Therefore, S_{ij} has the capacity to capture the pairwise correlations among cross-image feature pixels.

Any two images from the minibatch are fed into the old model $S_{\Theta^{t-1}}$ and new model S_{Θ^t} to obtain the feature pairs (e_i^{t-1}, e_j^{t-1}) and (e_i^t, e_j^t). Afterward, the feature pairs are, respectively, substituted into formula (5) to obtain the pairwise similarity matrix S_{ij}^{t-1} and S_{ij}^t. Then, the proposed pixel-to-pixel CFRD loss of every two images in the minibatch can be expressed

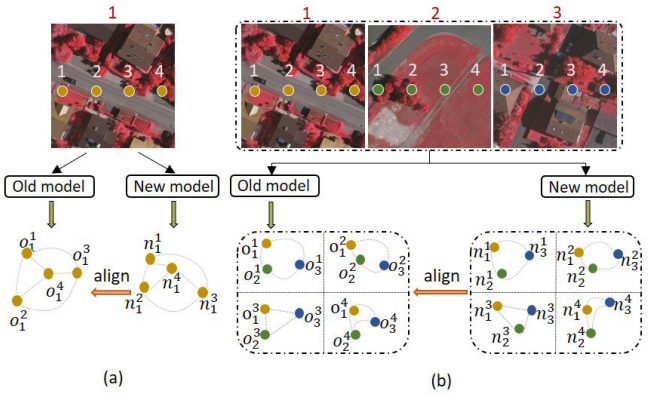


Fig. 4. Overview of modeling semantic relationship dependencies among pixels between a single image and across images. (a) Single-image feature relationship KD. (b) Cross-image feature relationship KD. Solid circles • with the same color represent feature pixels from the same image. O_i^j and N_i^j express the jth feature pixel of the ith image of the old model and the new model, respectively. The dotted line (- - -) means the similar relationship between the two feature pixels. The solid circles and dashed lines form a feature relational graph. Best viewed in color.

$$\mathcal{L}_{\text{cfrd}} = \frac{1}{N^2}\sum_{i=1}^{N}\sum_{j=1}^{N}\mathcal{L}_{p2p}\left(S_{ij}^t S_{ij}^{t-1}\right) \tag{6}$$

where $\mathcal{L}_{p2p}$ is formulated as

$$\mathcal{L}_{p2p}\left(S_{ij}^t, S_{ij}^{t-1}\right) = \frac{1}{A_f}\sum_{r=1}^{H_f}\text{KL}\left(\sigma\left(\frac{S_{ij|r,:}^t}{\tau}\right), \sigma\left(\frac{S_{ij|r,:}^{t-1}}{\tau}\right)\right) \tag{7}$$

where $S_{ij|r,:}^t$ means the rth row vector of S_{ij}, σ denotes the softmax activation function, and τ is the temperature coefficient employed to adjust the probability distribution. Kullback-Leibler (KL) denotes KL divergence, which is employed to constrain the pairwise similarity of the current model to be similar to that of the old model. Modeling the global semantic relationship between feature pixels in a minibatch effectively addresses the underutilization of features by existing CISS methods and facilitates the learning of a well-structured feature space. In addition, constraining S_{ij}^t to align with S_{ij}^{t-1} is beneficial for transferring similarity distributions among cross-image pixel embeddings, thereby allowing better retention of previous knowledge and reduction of CF.

D. CIFC Loss

Modeling cross-image feature pixel correlations is beneficial for constructing well-structured feature spaces. By distilling the output probability distribution and feature pixel correlation, and training with a CE loss (or its variants), the learned model can achieve a good rigidity-elasticity balance. As illustrated in Section I, features learned in this manner have good separability among classes, and the intraclass features are discrete rather than cohesive, as shown in Fig. 5. Inspired by Wen et al. [76],

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we add a variant of Centerloss for CISS dubbed CIFC loss to effectively increase the discriminative ability of the deeply learned features in neural networks. The CIFC loss can reduce the distance between features and feature centers of each class, thus making the intraclass distance more cohesive. Here, the features of each class are averaged to determine the feature center for that class.

However, CISS only contains the annotations of classes C^t at step t, and the remaining classes (the learned classes $C^{1:t-1}$ and the classes will be learned $C^{t+1:T}$) are processed as background class c_{bg}. Therefore, we draw support from the idea of pseudolabels, which are commonly used in the domain adaptation field, to recover the pixels' label of the learned category from the background class. Specifically, to identify whether a pixel belongs to the background or not for the segmentation GT map, we first create a binary mask M_{bg}

$$M_{\text{bg}} = \mathbb{I}\left(y_i^t(h,w) \in c_{\text{bg}}\right) \tag{8}$$

where $\mathbb{I}$ represents the indicator function. Then, the ith extended segmentation map (ESM) $\bar{y}_i^t$ obtained by the pseudolabel strategy at step t can be expressed as

$$\bar{y}_i^t(h,w) = \begin{cases} y_i^t(h,w), & \text{if } M_{\text{bg}} = 0 \\ \text{argmax}\, P_i^{t-1}(h,w), & \text{otherwise} \end{cases} \tag{9}$$

where P_i^{t-1} is the predicted segmentation map of the old model $S_{\Theta^{t-1}}$ and $\text{argmax}\, P_i^{t-1}(h,w)$ denotes the class label index of pixel (h,w). In other words, we retain the pixel labels that belong to the nonbackground class in the present step, while, for the background pixels, we replace them with the class label index that corresponds to the output probability of the previous model.

By downsampling the ESM obtained in Formula (9), we can acquire supervised label maps $\hat{y}^t \in \mathcal{R}^{H_f \times W_f}$ of the same size as the feature. It is important to note that we do not update the center for the entire training set but on a minibatch. Similar to the CFRD loss, we suppose that the size of training minibatch is N, and the network can obtain N project features $\{z_i \in \mathcal{R}^{H_f \times W_f \times D}\}|_{i=1}^N$ through the project head $g(\cdot)$. Here, H_f, W_f, and $D = 256\text{-}d$ are the spatial dimension and the channel dimension of z_i, respectively. We reshape the spatial dimension from $\{z_i \in \mathcal{R}^{H_f \times W_f \times D}\}|_{i=1}^N$, $\hat{y}^t \in \mathcal{R}^{H_f \times W_f}$ to $\{z_i \in \mathcal{R}^{A_f \times D}\}|_{i=1}^N$, and $\hat{y}^t \in \mathcal{R}^{A_f}$, where $A_f = H_f \times W_f$.

Since the old model learns the old classes' features better than the new model, we use the corresponding new and old project features when calculating the class center of the new and old classes, respectively. Consequently, the class center of each category at step t can be attained by the following formula:

$$C_{ct}^t[c] = \frac{1}{v} \sum_{i=1}^{N} \sum_{w=1}^{W_f} \sum_{h=1}^{H_f} \begin{cases} z_i^{t-1}\left[\hat{y}_i^t == c\right], & c \in C^{1:t-1} \\ z_i^t\left[\hat{y}_i^t == c\right], & c \in C^t \end{cases} \tag{10}$$

where v denotes the quantity of project pixels labeled as category c in the minibatch and $\hat{y}^t \in \mathcal{R}^{A_f}$ is the supervised label maps of features. $C_{ct}^t \in \mathcal{R}^{\mathcal{Y}^t \times D}$, where $\mathcal{Y}^t = C^{1:t}$ represents classes seen over time. Ultimately, the CIFC loss

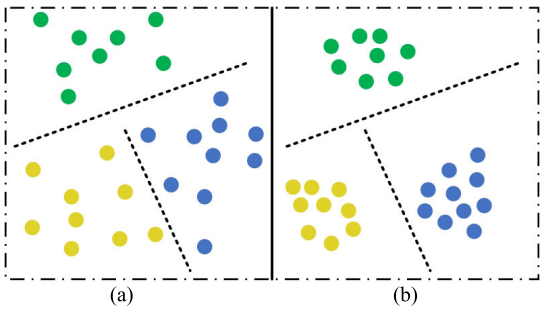


Fig. 5. Illustration of deeply learned features. Features learned by existing CIL methods are interclass separable but intraclass discrete. We want the network to learn interclass separable and intraclass cohesive features to better resist CF. (a) Discrete features. (b) Cohesive features. Best viewed in color.

can be expressed as

$$\mathcal{L}_{\text{cifc}} = \frac{1}{N_{\text{fg}}} \sum_{c=1}^{\mathcal{Y}^t} \sum_{i=1}^{N} \left(1. - \cos\left(z_i^t\left[\hat{y}_i^t == c\right], C_{ct}^t[c]\right)\right) \tag{11}$$

where $\cos(\cdot)$ denotes the cosine similarity function and N_{fg} indicates the number of foreground pixels of $\hat{y}^t$. The closer the class feature pixels are to the feature center, the greater the value of cosine similarity and the smaller the corresponding loss. Therefore, the proposed CIFC loss is beneficial to narrow the distance among the feature pixels belonging to the same class, which can achieve the purpose of gathering intraclass features. The cohesive intraclass features are more conducive to resisting CF.

E. CCWCE Loss

RS images possess the traits of large discrepancies in object size. If the object of each category is learned indiscriminately, the model is more inclined to focus on learning large-sized objects and ignore small objects. In order to increase the attention to small objects in the training process and, thus, improve the model's overall performance, we propose a CCWCE loss. Considering the irregularity of the objects in ISPRS datasets, we model the proportion of pixels across images to generate the corresponding class weights. Moreover, considering that the proportion of pixels in a minibatch is more stable than a single image, we use a minibatch as a unit to obtain the ratio of each category. Specifically, we first determine the class area by counting the quantity of GT labels for each category in the minibatch. Then, the pixel proportion of each category $C_p[c]$ in the minibatch is calculated by dividing the overall number of pixels, which can be expressed as

$$C_p[c] = \frac{1}{N \times H \times W} \sum_{i=1}^{N} \sum_{w=1}^{W} \sum_{h=1}^{H} \mathbb{I}\left(y_i^t(h,w) == c\right), \quad c \in C^t \tag{12}$$

where N is the size of minibatch and $\mathbb{I}$ denotes the indicator function.

According to formula (12), the ratio range of each category is [0, 1]. Therefore, we construct a monotonically declining function about C_p whose value is bigger than 1. We assign a larger weight to the category with a small proportion of pixels in the minibatch and a smaller weight to the category with a

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high proportion, which is beneficial to enhance the attention of the categories with few pixels and reduce the attention of the categories with more pixels. We consider the following three forms of weighted category pixel proportion:

$$\mathcal{W}_{\text{cls}} = \begin{cases} 1. + e^{-C_p} \\ 1. + e^{-2C_p} \\ 2. - C_p. \end{cases} \tag{13}$$

Taking the problem of background class semantic shift into account, we revisit the CE loss of equation (1) as [2]

$$\mathcal{L}_{\text{ce}}\left(x_i^t, y_i^t\right) = -\sum_{c \in C^t} y_i^t[c] \cdot \log\left(\tilde{P}\left(x_i^t; \Theta^t\right)[c]\right) \tag{14}$$

where

$$\tilde{P}\left(x_i^t; \Theta^t\right)[c] = \begin{cases} P\left(x_i^t; \Theta^t\right)[c] & \text{if } c \neq c_{\text{bg}} \\ \sum\limits_{k \in C^{1:t-1}} P\left(x_i^t; \Theta^t\right)[k], & \text{if } c = c_{\text{bg}}. \end{cases} \tag{15}$$

Then, the proposed CCWCE loss can be expressed as

$$\mathcal{L}_{\text{ccwce}} = \mathcal{W}_{\text{cls}} \cdot \mathcal{L}_{\text{ce}}. \tag{16}$$

IV. Experiments

We present the experimental setup in Section IV-A, including the datasets, evaluation criteria, and training details. Then, the standard CIL baselines are introduced in Section IV-B. In Section IV-C, we evaluate and analyze the ablation results of each proposed component in the most challenging setting. Finally, we contrast our framework with SOTA approaches from quantitative and qualitative aspects to further verify the efficacy of MiCro in Sections IV-D and IV-E.

A. Datasets, Metrics, and Training Details

We evaluate the efficacy of the proposed MiCro on the ISPRS 2D semantic labeling contest data, which consists of two datasets from two German cities: Vaihingen [21] and Potsdam [22].

1) ISPRS Vaihingen [21]: The Vaihingen 2D semantic labeling dataset consists of 33 patches, each of which measures 2494 × 2.064 pixels on average. Each patch contains a true orthophoto (TOP) obtained from a bigger TOP mosaic with its corresponding normalized digital surface model (nDSM) data. These tiles are densely annotated into six categories, which are arranged alphabetically as follows: 0: background (clutter); 1: building; 2: car; 3: impervious surface; 4: low vegetation; and 5: tree. We adopt 16 images defined by the organizer for training and the remaining 17 for testing.

2) ISPRS Potsdam [22]: The Potsdam 2D dataset includes 38 patches with the same shape of 6000 × 6.000 pixels, and the ground sampling distance is 5 cm. This dataset offers multiple data types, including not only red (R), green(G), blue (B), and near-infrared (NIR) channels but also DSM and normalized DSM (nDSM) data. Notably, we neither employ DSM nor nDSM data in all experiments. The organizer divided 24 of the 38 patches into the training set and the remaining 14 into the test set. Like the Vaihingen dataset, Potsdam is also intensively labeled into six categories: 0: background (clutter); 1: building; 2: car; 3: impervious surface; 4: low vegetation; and 5: tree (in alphabetical order).

3) iSAID [23]: The iSAID dataset is a semantic segmentation dataset for DOTA [77], comprising 2806 aerial images that have resolutions ranging from 800 × 800 to 4000 × 13.000 pixels. Among these 2806 images, 1411 images are the training set, 458 images are the validation set, and 957 images are the test set. This dataset contains a total of 655 451 instances that are spread across 15 distinct categories of objects, with the remaining pixels serving as the background. The 16 categories are listed in alphabetical order as follows: 0: background; 1: baseball diamond; 2: basketball court; 3: bridge; 4: ground track field; 5: harbor; 6: helicopter; 7: large vehicle; 8: plane; 9: roundabout; 10: ship; 11: small vehicle; 12: soccer ball field; 13: storage tank; 14: swimming pool; and 15: tennis court. Since the labels in the test set are not available, 20% of the training set is used for validation, and the validation set is used for the test.

4) Metrics: We adopt the mean intersection over union (mIoU) to reasonably assess all methods' performance. Suppose that $\text{mIoU}_{t,k} (k \leq t)$ represents the precision of the kth step after learning from step 1 to t, which can be described as follows:

$$\text{mIoU}_{t,k} = \frac{1}{|C^k|} \sum_{i=1}^{|C^k|} \frac{p_{ii}}{\sum_{j=1}^{C^k} p_{ij} + \sum_{j=1}^{C^k} p_{ji} - p_{ii}} \tag{17}$$

where C^k indicates the amount of categories in the kth step and p_{ij} denotes the number of pixels that belong to category i but are identified by the network as category j.

5) Implementation Details: All experiments are performed with half-precision for efficiency reasons on two NVIDIA TITAN RTX GPUs with 24-GB memory. In addition, all approaches are implemented using Pytorch and evaluated with the same experimental setup. Concretely, all methods adopt the Deeplab-V3 architecture [72] with ResNet101 [71] as the backbone and an output stride of 16. The ImageNet pretrained model [78] is loaded to initialize the backbone only in the first task, and the models of subsequent tasks are initialized with the previous model. To train the network, we employ stochastic gradient descent (SGD) and a polynomial decay algorithm with a power of 0.9. The starting learning rate for each method is set to 0.01 in the initial training step and 0.001 for all succeeding steps. We randomly crop the 16 training patches from Vaihingen and the 24 patches from Potsdam into 8000 and 12 000 images of 512 × 512 pixels, respectively. Note that, during the training phase, we further split the training set into training and validation sets by 4:1 like in [2] and [3]. For the test set of the two datasets, we crop the patches to 512 × 512 pixels with a sliding window overlap of 128. For the iSAID dataset, we crop the images into a size of 896 × 896 pixels (all resized to 512 × 512 pixels before being sent to the network) with the overlap of the sliding window of 384. For data augmentation, we apply a random resize crop (from 0.5 to 2) and random horizontal flipping on all training data. For the three datasets used, we train the model 40 epochs with a batch size of 24 in each learning step. Referring to the

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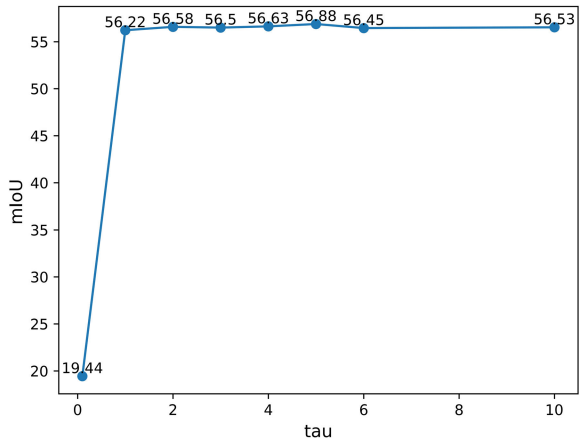


Fig. 6. Ablation results of different temperature τ studies for the proposed CFRD module on the 1s setting of the ISPRS Vaihingen dataset.

settings of [2] and [3], we set $\alpha = 100$ only for 1s incremental setting and $\alpha = 10$ for others. β is set to 10, and γ is 1 for all incremental settings. It is worth noting that we construct the incremental datasets D_{tr} by splitting the whole dataset into several disjoint image sets.

B. Baselines and Incremental Settings

1) Baselines: We report results for eight regularization-based methods (three prior-focused and five data-focused) on the Vaihingen and Potsdam datasets. Moreover, we explore two additional methods: fine-tuning (FT) and offline learning (Joint). Since Joint trains all data offline at the same time, it is generally considered to be the upper boundary on continual learning performance.

For the selection of prior-focused approaches, we follow [2] and adopt three methods of elastic weight consolidation (EWC) [54], synaptic intelligence (SI) [57], and Riemannian walk (RW) [58]. While LWF [55], iCaRL [41], ILT [1], MiB [2], and HCISS [3] are chosen for the data-focused approaches, iCaRL is the single-head implementation of LWF built on several binary classifiers. ILT, MiB, and HCISS are explicitly designed for CISS. On the basis of LWF, ILT additionally introduces IFD loss. MiB optimizes two LWF losses by considering the background. HCISS mitigates CF by exploiting historical information to recover pixels of learned classes from background classes and enhance the global perception of features.

Different from the above methods that only construct the relationship among pixels in a single image, this article focuses on exploiting the semantic dependencies among pixels across images in the current training minibatch.

2) Incremental Settings: Although Tasar et al. [70] also conduct experiments on the two ISPRS datasets, they only focus on an incremental setting with a certain order. In order to exclude the variation of results caused by different incremental orders, we perform incremental learning on the dataset categories in alphabetical order for all experiments. Considering that the pixels of the clutter category occupy a relatively modest portion of the whole dataset (0.78%), we treat it as the background category in all experiments for both datasets.

We hope to construct reasonable incremental settings for fair and unbiased comparisons with other methods. For the remaining five foreground categories (in alphabetical order) of the two ISPRS datasets, we divide four incremental settings to examine the efficacy and robustness of the proposed framework. These four incremental settings are 4-1 (two steps), 3-1s (three steps), 2-1s (four steps), and 1s (five steps). 4-1 denotes learning four classes followed by one sequentially. While 3-1s (2-1s) means to learn three (two) classes first and then learn one class at a time for two (three) increments, 1s refers to learning one class followed by four times one class, which is the most challenging setting of all incremental settings. As for the iSAID dataset, we use the incremental division way of [3] and conduct experiments on 10-5 (two steps), 5 s (three steps), and 10-1s (six steps) settings.

C. Ablation Study

Since the 1s setting contains the most incremental processes, the model is most prone to CF during the learning process, so it is regarded as the most challenging setting. We perform a series of ablation experiments on the 1s setting of the ISPRS Vaihingen dataset to verify the validity of the presented modules. Notably, the other three incremental settings are not discussed in the ablation experiments because the incremental steps are few, and the accuracy of many methods is near the upper bound of joint training. Unless especially emphasized, all experimental results in this article are in percentile.

1) Performance Evaluation of Different Components: We provide the contributions of the presented modules and the combinations among them in detail in Table I. The first row of Table I gives the outcomes of ILT [1], which uses CE loss, OPD loss, and IFD loss to combat CF. Note that the OPD loss is used in all combinations, so it will not be emphasized in the subsequent experimental analysis. We replace the IFD loss with the CFRD loss and get the results in the second row in Table I. By applying the CFRD loss, IoU is increased for each category, and the mIoU has grown by +5.93 percentage points (p.p) compared to the baseline (ILT). The improvement suggests that modeling cross-image pairwise feature similarity is more conducive to learning a well-structured feature space than simply constraining old and new model features to be similar in a single image and thus better resistant to CF during continual learning. Then, instead of using the CE loss in the subsequent incremental steps, we employ the proposed CCWCE loss to conduct experiments. The mIoU is +7.59 p.p greater than the baseline and +1.66 p.p higher than the combination of CFRD loss and CE loss, which illustrates the validity of CCWCE loss. In addition, we experiment with various combinations of the three losses that have been presented, including CE loss combined with CFRD loss and CIFC loss, CCWCE loss combined with CIFC loss, and CFRD loss combined with CCWCE loss and CIFC loss. The proposed three losses are combined in pairs, which cross-verifies the effectiveness of each component and proves that both losses are mutually beneficial. In particular, our method reaches the most advanced level when the three losses are applied simultaneously. It is worth noting that, for the category of

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TABLE I

ABLATION STUDIES OF DIFFERENT MODULES ON THE ISPRS VAIHINGEN TEST SET IN THE MOST CHALLENGING SETUP 1 s WITH FIVE STEPS. PER-CLASS IOU AND MEAN IOU ARE LISTED (%). THE BEST OUTCOMES ARE SHOWN IN BOLD, WHILE THE SUBOPTIMAL RESULTS ARE UNDERLINED

CE	OPD	FPD	CFRD	CCWCE	CIFC	**1s** (5 steps) Building	Car	Imp. Surf	Low Vege	Tree	all	Imprv.
√	√	√				85.43	0	68.69	37.30	63.33	50.95	0
√	√		√			86.19	1.86	74.47	52.87	69.01	56.88	+5.93
	√		√	√		85.84	10.96	74.37	53.36	68.16	58.54	+7.59
√	√		√		√	86.50	5.07	76.37	54.72	**71.02**	58.74	+7.79
	√			√	√	85.98	11.64	74.53	53.41	68.23	58.76	+7.81
	√		√	√	√	**86.84**	**21.80**	**76.77**	**56.63**	70.67	**62.54**	+11.59

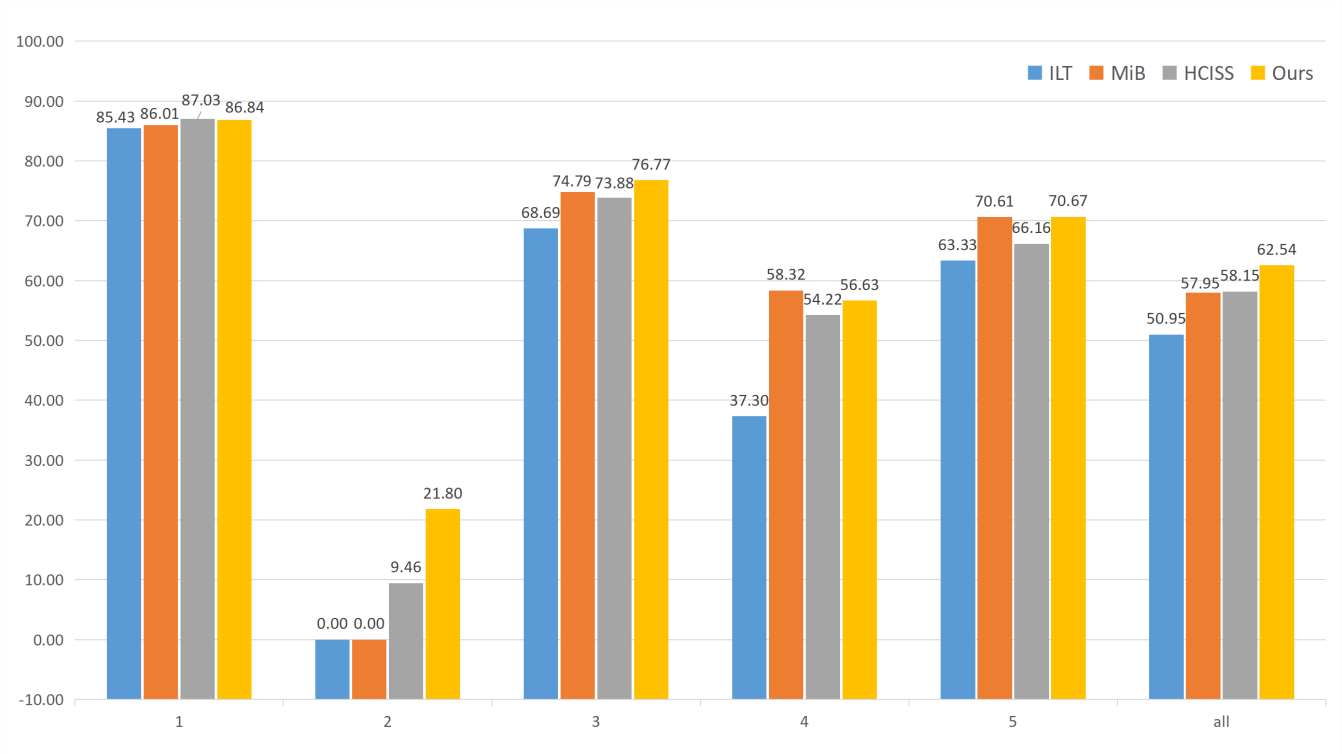


Fig. 7. Per-class IoU for each category on the most challenging setting 1s of the ISPRS Vaihingen dataset. Best viewed in color.

TABLE II

ABLATION RESULTS OF DIFFERENT CLASS WEIGHTING FUNCTIONS ω FOR THE CCWCE LOSS ON THE 1 s SETTING OF THE ISPRS VAIHINGEN DATASET. PER-CLASS IOU AND MEAN IOU ARE LISTED (%)

ω	**1s** (5 steps) Building	Car	Imp. Surf	Low Vege	Tree	all	Imprv.
base	**86.19**	1.86	**74.47**	52.87	**69.01**	56.88	0
ω_1	86.03	6.42	74.44	53.12	68.33	57.67	+0.79
ω_2	85.92	9.08	74.34	**53.32**	68.19	58.17	+1.29
ω_3	85.93	**12.03**	74.43	53.20	68.02	**58.72**	**+1.84**

TABLE III

ABLATION RESULTS OF VARIOUS HYPERPARAMETERS γ FOR THE CIFC MODULE ON THE 1 s SETTING OF THE ISPRS VAIHINGEN DATASET. PER-CLASS IOU AND MEAN IOU ARE LISTED (%)

γ	**1s** (5 steps) Building	Car	Imp. Surf	Low Vege	Tree	all	Imprv.
base	85.93	12.03	74.43	53.20	68.02	58.72	0
0.01	85.91	11.98	74.41	53.20	68.10	58.72	0
0.05	86.07	11.72	74.50	53.43	68.25	58.79	0.07
0.1	86.34	12.73	74.78	53.04	68.16	59.01	+0.29
0.5	86.54	14.79	75.53	55.34	69.63	60.37	+1.65
1	**86.84**	**21.80**	**76.77**	**56.63**	**70.67**	**62.54**	**+3.82**

car (small object with a small number of pixels), we increase its IoU from 0 of the baseline to 21.8%, and the overall mIoU is increased by +11.59 p.p. It can be observed that the CCWCE loss is particularly useful for the car class with

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TABLE IV

MEAN IoU ON THE ISPRS VAIHINGEN TEST SET FOR DIVERSE CIL APPROACHES. THE BEST OUTCOMES ARE SHOWN IN BOLD, WHILE THE SUBOPTIMAL RESULTS ARE UNDERLINED

Method	**4-1** (2 steps)			**3-1s** (3 steps)			**2-1s** (4 steps)			**1s** (5 steps)		
	1-4	5	all	1-3	4-5	all	1-2	3-5	all	1	4-5	all
FT	0	75.11	15.02	0	36.87	14.75	0	24.85	14.91	0	18.44	14.75
EWC [54]	0.03	74.31	14.88	0	35.61	14.25	0	21.52	12.91	0	14.32	11.46
SI [57]	0	74.99	15.00	0	37.00	14.80	0	24.83	14.90	0	18.49	14.79
RW [58]	0.03	74.59	14.94	0	35.07	14.03	0	21.49	12.89	0	12.60	10.08
LWF [55]	**72.65**	75.20	**73.16**	74.56	65.51	70.94	66.36	67.02	66.76	85.89	48.47	55.95
iCaRL [41]	72.37	74.68	72.83	**75.11**	35.00	59.07	65.71	20.75	38.73	83.22	13.53	27.47
ILT [1]	72.55	75.05	73.05	74.38	60.31	68.75	65.73	58.68	61.50	85.43	42.33	50.95
MiB [2]	72.59	75.00	73.07	74.30	68.23	71.87	70.76	70.09	70.36	86.01	50.93	57.95
HCISS [3]	72.50	**75.29**	73.06	74.19	**68.53**	71.93	**71.28**	69.61	70.28	**87.03**	50.93	58.15
MiCro	72.29	75.02	72.84	74.77	67.75	**71.96**	69.59	**70.97**	**70.42**	86.84	**56.42**	**62.54**
Joint	72.89	75.34	73.38	75.46	70.28	73.38	73.02	73.67	73.38	86.96	69.99	73.38

TABLE V

MEAN IoU ON THE ISPRS POTSDAM TEST SET FOR DIVERSE CIL APPROACHES. THE BEST OUTCOMES ARE SHOWN IN BOLD, WHILE THE SUBOPTIMAL RESULTS ARE UNDERLINED

Method	**4-1** (2 steps)			**3-1s** (3 steps)			**2-1s** (4 steps)			**1s** (5 steps)		
	1-4	5	all	1-3	4-5	all	1-2	3-5	all	1	4-5	all
FT	0	72.40	14.48	0	35.54	14.22	0	22.94	13.76	0	16.8	13.44
EWC [54]	0.07	70.30	14.11	0.01	33.61	13.45	0.01	20.75	12.45	0	15.07	12.06
SI [57]	0	72.21	14.44	0	35.95	14.38	0	23.08	13.85	0	16.76	13.41
RW [58]	0	70.43	14.09	0	33.59	13.44	0.10	20.68	12.45	0	14.61	11.69
LWF [55]	81.10	73.98	79.68	83.82	70.32	78.42	81.63	70.18	74.76	89.14	67.04	71.46
iCaRL [41]	**81.67**	66.29	78.60	84.01	28.84	61.96	79.60	17.00	42.04	88.49	11.62	26.59
ILT [1]	80.72	73.41	79.26	83.10	67.48	76.85	81.01	65.49	71.70	87.99	54.59	61.27
MiB [2]	80.92	73.01	79.34	83.42	70.28	78.17	83.08	70.06	75.27	88.49	67.83	71.96
HCISS [3]	81.02	73.01	79.41	83.56	70.82	78.47	**83.54**	70.11	75.49	89.28	67.22	71.64
MiCro	81.19	**74.11**	**79.77**	**84.03**	**70.88**	**78.77**	83.14	**72.57**	**76.80**	**89.43**	**70.69**	**74.44**
Joint	81.55	73.83	80.01	84.61	73.10	80.01	85.8	76.15	80.01	90.80	77.31	80.01

TABLE VI

MEAN IoU ON THE ISPRS POTSDAM DATASET FOR DIVERSE CISS APPROACHES IN THE MOST CHALLENGING SETUP OF 1 S

Method/class	**0** **BG**	**1** (1st step) **Building**	**2** (2nd step) **Car**	**3** (3rd step) **Imp. Surf**	**4** (4th step) **Low Vege**	**5** (5th step) **Tree**	**mIoU FG**	**mIoU FG+BG**
LWF [55]	19.45	89.14	58.56	76.68	65.42	67.51	71.46	62.79
ILT [1]	14.04	87.99	27.09	69.73	59.45	62.08	61.27	53.40
MiB [2]	12.56	88.49	63.48	76.13	65.73	65.97	71.96	62.06
HCISS [3]	20.89	89.28	65.30	76.74	64.48	62.38	71.64	63.18
MiCro	**22.51**	**89.43**	**70.14**	**77.87**	**66.44**	**68.31**	**74.44**	**65.79**
Joint	32.93	90.80	80.80	82.23	72.38	73.83	80.01	72.16

a small proportion of pixels, while the CFRD and CIFC are not specific to a particular case but are valid for all classes.

2) Sensitivity Analysis of the Temperature τ for the CFRD Loss: In the proposed CFRD loss, we apply formula (7) to calculate the pixel-to-pixel CFRD loss of every two images in minibatch. The temperature coefficient τ is employed to adjust the probability distribution of the pixel-to-pixel similarity matrix between the new and old models. We present the outcomes of the different τ values in Fig. 6. It can be observed that, when τ is less than 1 (the first point, where $\tau = 0.1$), the probability distribution becomes sharp, which can adversely affect model learning. When the temperature coefficient τ is between 1 and 10 ($\tau = 1$, 2, 3, 4, 5, 6, and 10), the numerical change has little effect on the result, and the mIoU is stable

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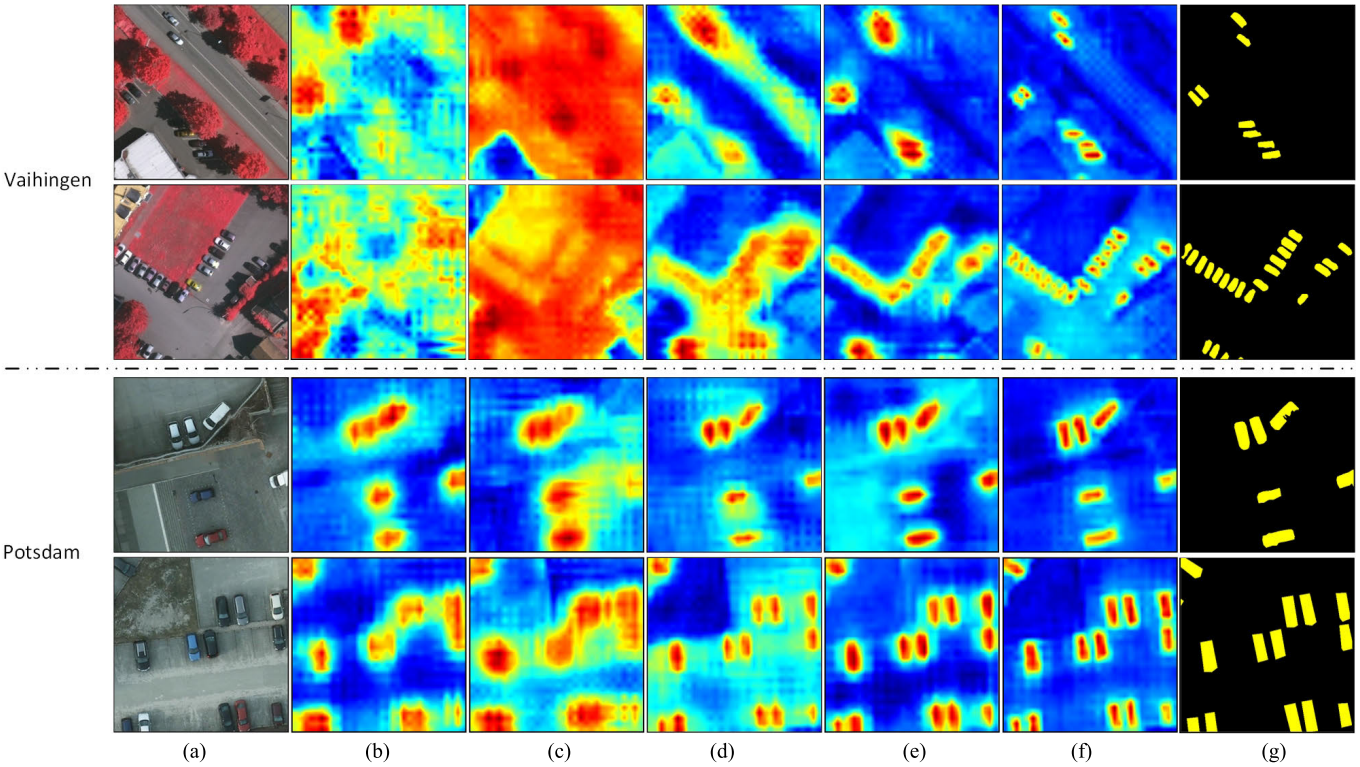


Fig. 8. Feature visualization results from different methods in the car class. It can be observed that our method has a stronger learning ability for small objects than other SOTA methods, and the learned object boundary is clearer, which is close to the level of joint training. (a) Image. (b) ILT. (c) MiB. (d) HCISS. (e) Ours. (f) Joint. (g) Car_GT. Best viewed in color.

at around 56%. The value of τ is set to 5 in all subsequent experiments.

3) Evaluation of Different Weighting Functions ω for the CCWCE Loss: As described in formula (13), we consider three forms of weighted category pixel proportion and verify their effect on the 1s setting of the ISPRS Vaihingen dataset. Table II provides the ablation outcomes, where $\omega_1 = 1. + e^{-C_p}$, $\omega_2 = 1. + e^{-2C_p}$, and $\omega_3 = 2. - C_p$. According to the table, the linear function ω_3 has a better impact than the exponential functions ω_1 and ω_2. Note that base refers to the result obtained after CFRD loss replaces IFD loss in ILT [1]. The results indicate that reversely assigning the corresponding class weights to the proportion of pixels in the minibatch is beneficial to the learning of small objects. The IoU of the car category increases from 1.86% to 12.03%.

4) Evaluation of Different Hyperparameter γ for the CIFC Loss: For the two hyperparameters α and β that applied to the OPD loss and CFRD loss in formula (4), we refer to the [3] to take 10 (100 only for the 1s setting) and 10, respectively. The CIFC loss is used together with CCWCE loss to control the cohesiveness of various intraclass features. We perform a set of ablation experiments for the hyperparameter γ, and the results are presented in Table III. When the value of γ is small ($\gamma = 0.01$ or 0.05), it can be seen that its impact on the overall results is negligible. As the value of γ increases and gradually approaches 1, the performance of both a single class and the whole is improved. Particularly, the model performance reaches the SOTA level when $\gamma = 1$, and the mIoU is 62.54%. Note that the base here refers to the result obtained by replacing IFD and CE losses with CFRD and CCWCE losses, respectively.

D. Quantitative Analysis

We contrast MiCro with the existing SOTA approaches on the four incremental settings of the two ISPRS datasets.

1) ISPRS Vaihingen: Table IV reveals the performance of the proposed MiCro framework and eight regularization-based approaches on the ISPRS Vaihingen dataset. On the four incremental settings, the three prior-focused methods (EWC, SI, and RW) and the FT approach are effective in learning new classes, but they have suffered serious CF for the previously learned classes. In the first three incremental settings, the performance of the methods in the first four rows of Table IV is nearly 55 p.p lower than that of the following six data-focused approaches on average. In the most challenging 1s setting, the learning effects of the three prior-focused methods and FT are worse, with an average mIoU of 13%.

Since existing SOTA approaches are all very near to the performance upper bound and have little space for improvement in the 4-1 setting, our MiCro does not demonstrate an advantage over this simple setup. For the 3-1s setting, it can be observed that the mIoUs of LWF, MiB, HCISS, and MiCro all exceed 70%. The performance of MiB, HCISS, and MiCro is even close to 72%, which is a small difference from the performance upper bound. On the 3-1s and 2-1s settings, MiCro performs slightly better than MiB and HCISS, with mIoU of 71.96% and 70.42%, reaching SOTA. On the 1s setting with the most incremental steps, MiCro achieves a

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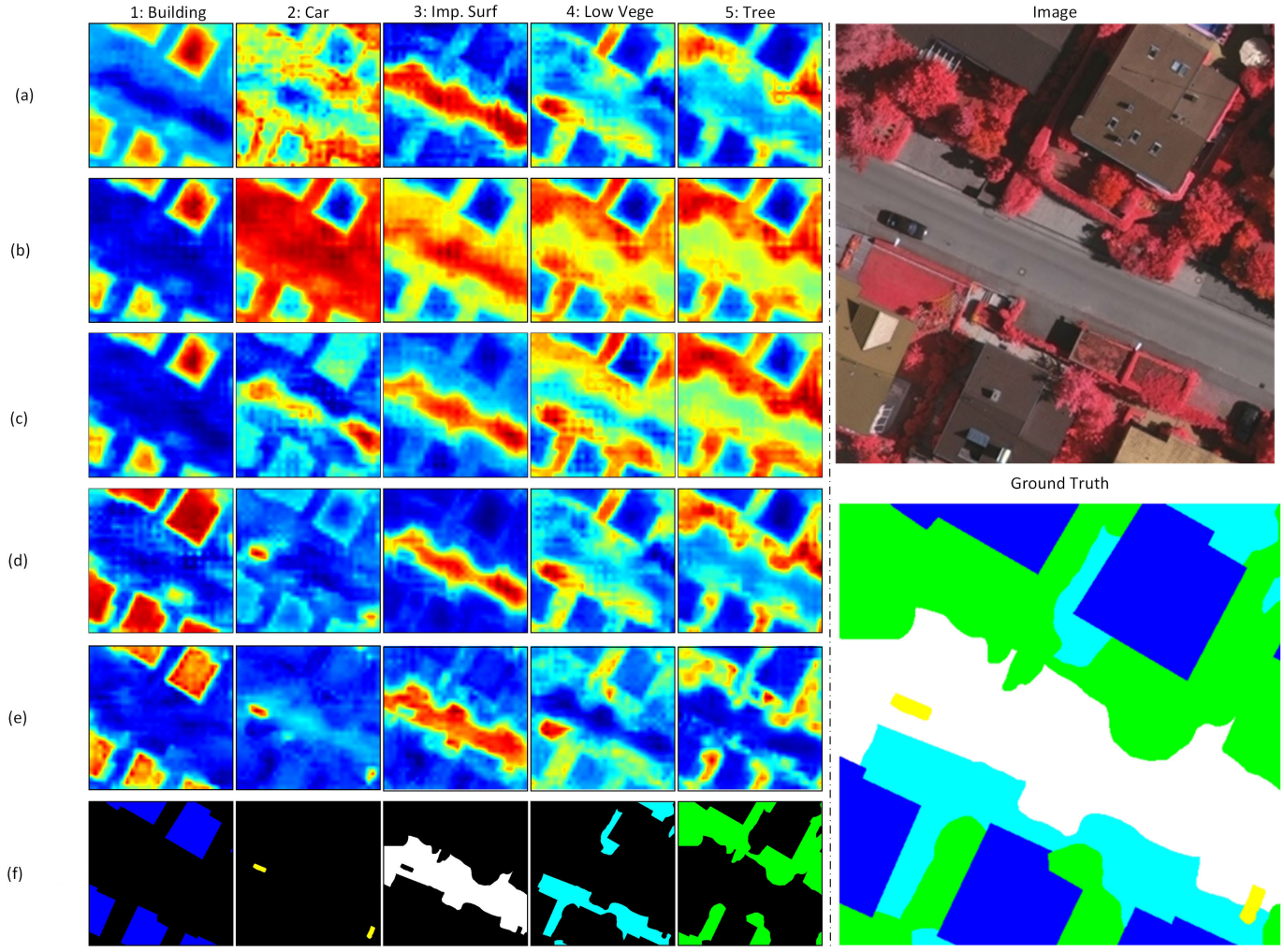


Fig. 9. Visualization results of CAMs for each category on the ISPRS Vaihingen dataset on the most challenging setting 1s. Note that rows indicate methods and columns denote categories. (a) ILT. (b) MiB. (c) HCISS. (d) Ours. (e) Joint. (f) CLS_GT expresses the GT of each category in the complete GT in the right figure. Best viewed in color.

superior performance of 62.54%, surpassing ILT by +11.59 p.p, MiB by +4.59 p.p, and HCISS by +4.39 p.p. In Fig. 7, we present the IoU results of SOTA methods and MiCro on each category of ISPRS Vaihingen dataset 1s setting. It can be seen that MiCro is close to or reaches the SOTA level in all categories. For the second category (Car), both ILT and MiB fail to segment. Only HCISS and MiCro are valuable, and MiCro is much higher than HCISS.

2) ISPRS Potsdam: Table V shows experiments on Potsdam 4-1, 3-1s, 2-1s, and 1s settings. It can be observed that MiCro achieves SOTA on all incremental settings. The performance of EWC, SI, RW, and FT is consistent with the trend in the Vaihingen dataset, and their performance is much lower than the data-focused approaches. In all incremental settings, they have the ability to learn new classes but completely forget old ones. Similarly, in the simplest setting of 4-1, the data-focused approaches only produce slight forgetting, and the overall performance approximates the Joint training. The proposed MiCro achieves mIoU of 79.77% on 4-1, which is only 0.24 p.p less than the upper bound of 80.01%. It is worth mentioning that MiCro even surpasses the Joint training in learning the new class (the fifth class) by +0.28 p.p. On the task of 3-1s with two incremental steps, MiCro still outperforms other SOTA methods. The mIoU of MiCro is more than +1 p.p higher than the SOTA methods, especially in the new classes on the 2-1s setup. In addition to the results on the 1s setting given in Table V, we also report the performance of each category and their overall performance (including the background category) in detail in Table VI. It can be observed that MiCro not only surpasses the existing SOTA methods in the overall mIoU but also reaches the leading level for single-category learning and knowledge preservation. Only concentrating on the foreground category, MiCro's mIoU is +13.17 p.p higher than baseline (ILT), +2.48 p.p higher than MiB, and +2.80 p.p higher than HCISS.

3) iSAID: To test the universality of the proposed method, we also conduct experiments on the iSAID dataset, and the results are shown in Table VII. It can be observed that MiCro reaches the SOTA level on all three incremental settings. On the 10-5 setting, MiCro achieves an mIoU of 56.90 in the new class, which is less than 1 p.p lower than Joint training. In the last two steps of 5 s, MiCro's results are much higher

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TABLE VII

MEAN IOU ON THE iSAID TEST SET FOR DIFFERENT CIL METHODS. THE BEST OUTCOMES ARE SHOWN IN BOLD, WHILE THE SUBOPTIMAL RESULTS ARE UNDERLINED. † REPRESENTS RESULTS EXCERPTED FROM [3]

Method	**10-5** (2 steps)			**5s** (3 steps)				**10-1s** (6 steps)		
	1-10	10-15	all	1-5	5-10	11-15	all	1-10	11-15	all
FT†	0.00	51.10	17.03	0.00	0.00	47.64	15.88	0.00	15.63	5.21
LWF† [55]	44.70	45.41	44.94	36.53	26.70	38.44	33.89	4.99	16.81	8.93
iCaRL† [41]	50.96	24.11	42.01	42.08	5.20	23.00	23.42	12.19	6.09	10.16
ILT† [1]	49.25	35.14	44.55	40.80	9.68	31.30	27.26	15.58	19.46	16.87
MiB† [2]	47.80	47.34	47.65	**46.75**	17.83	37.54	34.04	25.06	22.94	24.35
HCISS† [3]	**52.52**	39.52	48.19	43.54	23.94	35.67	34.39	**31.48**	27.65	30.21
Micro	49.17	**56.90**	**51.75**	44.09	**43.22**	**50.28**	**45.86**	28.82	**38.01**	**31.88**
Joint†	58.81	57.88	58.50	57.63	59.98	57.88	58.50	58.81	57.88	58.50

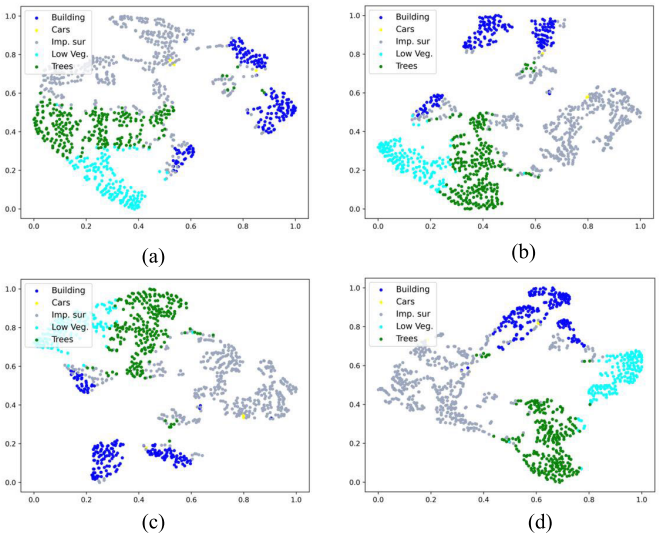


Fig. 10. T-SNE of feature embeddings on the ISPRS Vaihingen test set for four methods. We can observe that the class features learned by our proposed method are more cohesive compared to the existing SOTA methods. (a) ILT. (b) MiB. (c) HCISS. (d) Ours. Best viewed in color.

than the existing CIL methods, and the overall mIoU on 5 s exceeds SOTA by +11.67 p.p and baseline by +18.60 p.p. On the most difficult 10-1s incremental setup, MiCro is also optimal, exceeding the baseline by +15.01 p.p.

Quantitative results on the three datasets illustrate that, compared with the existing SOTA methods that only consider aligning old and new models and learning new knowledge on a single image, accounting for pixel semantic dependencies based on minibatch images is more beneficial for maintaining the balance of elasticity stability in incremental processes.

E. Qualitative Analysis

In the ablation analysis of Section IV-C, we discuss the validity of MiCro on small objects through experimental data. Meanwhile, the experimental findings also confirm the validity of the proposed CIFC loss in making intraclass features more cohesive. Although the data can reflect the final results, it is not intuitive. Therefore, we generate some visualizations on the most challenging setting 1s for intuitive display in this section. First, we visualize the features of the existing SOTA methods and MiCro with t-stochastic neighbor embedding (T-SNE). Second, we do feature visualization for small objects on both used datasets. Furthermore, we also visualize the class activation maps (CAMs) on the ISPRS Vaihingen dataset. Finally, we present the segmentation visualization for different approaches on the 1s setting of the ISPRS Potsdam dataset.

1) T-SNE Visualization Analysis: Fig. 10 shows the feature dimension reduction visualization results of the existing SOTA methods and MiCro. It can be seen that, although the SOTA methods can well separate the features of each class, the intraclass feature pixels are too scattered. For example, the blue feature pixels learned by ILT, MiB, and HCISS are divided into three regions, while the features learned by the proposed MiCro are concentrated in one region. The T-SNE results again prove the effectiveness of the CIFC loss for narrowing the intraclass distance from a visual perspective.

2) Small Objects' Feature Visualization Analysis: Fig. 8 shows the results of feature visualizations for small objects (Car) on two used datasets. It can be observed from columns 2 and 3 that ILT and MiB are completely unable to correctly distinguish the car class on the Vaihingen dataset, and the entire region is misactivated. The features learned on the Potsdam dataset are also blurred and indistinguishable. HCISS works much better than ILT and MiB. It can distinguish the boundary of the object on the Potsdam dataset, but there are still false activations of the surrounding area. Whether on Vaihingen or Potsdam datasets, the proposed MiCro can learn small objects' features that are far more effective than the SOTA approaches. The boundary among the object features learned by MiCro is relatively clear, and the area of false activation is especially small, which is more similar to the features obtained by Joint training. Notably, the results in the second line also reveal that our method has a good learning ability for densely arranged objects in RS scenes.

3) CAMs' Visualization Analysis: We show the CAMs of SOTA methods and MiCro on the ISPRS Vaihingen dataset in Fig. 9. Each column corresponds to a foreground category (note that the categories are in alphabetical order). For the first class (column), all the methods work well. However,

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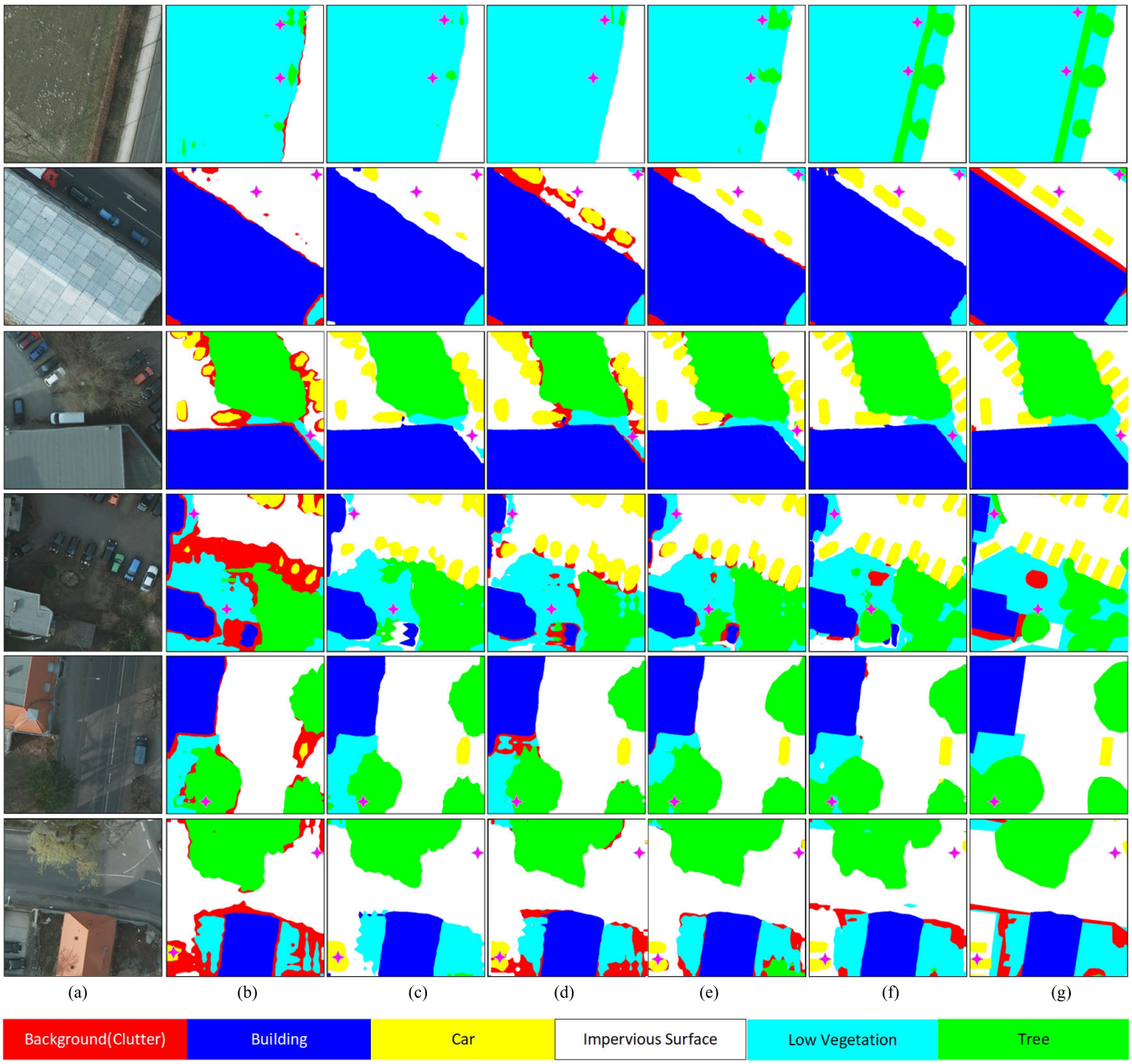


Fig. 11. Visualization of segmentation results of different methods on the most challenging setting 1s on the ISPRS Potsdam dataset. Purple quadrangular stars are the focus area. (a) Images. (b) ILT. (c) MiB. (d) HCISS. (e) Ours. (f) Joint. (g) GT. Best viewed in color.

ILT and MiB have extensive false activation in the second category. In contrast, HCISS activates the object area more precisely, but inaccurate positioning still exists. Our MiCro correctly activates the object area. The object's shape is clear, and the effect is similar to Joint training. Similarly, the SOTA methods either have the problem of insufficient or excessive activation of the object area in the following three categories, while the visualization results of MiCro are still close to Joint training. It can be seen that modeling cross-image features are conducive to learning discriminant features in each category and help to resist CF in the incremental process.

4) Segmentation Results Visualization Analysis: We visualize the segmentation results of SOTA methods and MiCro on the 1s setting of the Potsdam dataset in Fig. 11. As can be observed from the results in the first and fifth rows, SOTA approaches are highly missegmented for the tree category, while MiCro's results are significantly better than theirs. The results in rows 2–4 show that SOTA methods suffer from severe undersegmentation for the categories car and low vegetation, as indicated by the purple quadrangular stars, while MiCro's results are close to the Joint upper bound. It can also be observed in the last row that, in addition to the car category, MiCro has a better segmentation effect for the background class (clutter). In general, MiCro outperforms the existing SOTA methods, as evidenced by better segmentation of each class, less missed segmentation, and clearer segmentation boundaries. These results adequately illustrate the importance of learning more discriminative features for CISS.

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V. Conclusion

This article focuses on the CISS paradigm in RS images. To learn a feature space capable of overcoming CF, we propose an effective and generalized framework for MiCro. MiCro mainly mines the correlation of pixels between different images in minibatch to construct an interclass separable, intraclass cohesive, and well-structured feature space. In particular, we propose CFRD loss to transfer the similarity distributions among cross-image pixel embeddings. In addition, CCWCE loss is devised to enhance the model's attention to small objects and categories with few pixels in the image. CIFC loss is presented to learn interclass separable and intraclass cohesive feature space with CCWCE. Extensive experiments on multiple incremental settings that outperform the SOTA methods verify the efficacy of the MiCro framework. We hope that the open-source code can promote the development of CISS in the field of RS. As for future work, we may consider combining CISS with semisupervised learning because the incompletely labeled GT given by CISS in the incremental process can also be regarded as a semisupervised type.

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